

TOPICS IN MATHEMATICAL SCIENCE

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INTRODUCTION TO KOSZUL DUALITY

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Convention

Throughout the course, $\mathbb{k}$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\mathbb{k}$ -vector spaces.

1 Lecture 1: Graded rings and modules

Algebras

We begin with the basic object of the course.

Definition 1.1. A $\mathbb{k}$ -algebra is a ring A together with a ring homomorphism

$$\mathbb{k} \longrightarrow Z(A),$$

where $Z(A) = \{a \in A \mid ab = ba \text{ for all } b \in A\}$ is the center of A . Equivalently, A is a $\mathbb{k}$ -vector space with an associative multiplication

$$A \times A \rightarrow A, \quad (a, b) \mapsto ab,$$

which is $\mathbb{k}$ -bilinear and has an identity element 1_A .

The equivalence in the definition is useful. The ring point of view remembers multiplication; the vector space point of view makes clear that multiplication distributes over linear combinations.

Example 1.2. (1) The field $\mathbb{k}$ is a $\mathbb{k}$ -algebra.

(2) The polynomial ring $\mathbb{k}[x]$ is a $\mathbb{k}$ -algebra.

(3) The matrix ring $\text{Mat}_n(\mathbb{k})$ is a $\mathbb{k}$ -algebra. It is usually non-commutative when $n \geq 2$.

(4) If V is a $\mathbb{k}$ -vector space, then $\text{End}_{\mathbb{k}}(V)$ is a $\mathbb{k}$ -algebra under composition.

(5) The quotient $\mathbb{k}[x]/(x^2)$ is a $\mathbb{k}$ -algebra. It has basis $\{1, x\}$ and relation $x^2 = 0$.

Definition 1.3. Let A be a $\mathbb{k}$ -algebra. A right A -module is a $\mathbb{k}$ -vector space M together with a map

$$M \times A \rightarrow M, \quad (m, a) \mapsto ma$$

such that, for all $m, n \in M$, $a, b \in A$, and $\lambda \in \mathbb{k}$,

$$(m + n)a = ma + na, \quad m(a + b) = ma + mb, \quad (\lambda m)a = \lambda(ma),$$

$$m(ab) = (ma)b, \quad m1_A = m.$$

Thus a module is a vector space on which elements of A act as linear operators, compatibly with multiplication in A . For instance, a module over $\mathbb{k}[x]$ is the same thing as a vector space M together with one linear operator $T : M \rightarrow M$, where $mx := T(m)$.

Graded algebras

Koszul duality is a graded theory. The grading records a notion of degree, and the main results compare algebraic multiplication with homological degree.

Definition 1.4. A $\mathbb{Z}_{\geq 0}$ -graded $\mathbb{k}$ -algebra is a $\mathbb{k}$ -algebra A with a direct sum decomposition

$$A = \bigoplus_{i \geq 0} A_i$$

as $\mathbb{k}$ -vector spaces such that

$$A_i A_j \subset A_{i+j}$$

for all $i, j \geq 0$, and $1_A \in A_0$. An element $a \in A_i$ is called *homogeneous of degree i* .

Every element $a \in A$ is a finite sum

$$a = a_0 + a_1 + \cdots + a_N, \quad a_i \in A_i.$$

The summands a_i are called the *homogeneous components* of a .

Example 1.5. The polynomial ring $\mathbb{k}[x]$ is graded by degree:

$$\mathbb{k}[x] = \bigoplus_{i \geq 0} \mathbb{k}x^i.$$

Here $A_i = \mathbb{k}x^i$. More generally,

$$\mathbb{k}[x_1, \dots, x_n] = \bigoplus_{i \geq 0} \mathbb{k}[x_1, \dots, x_n]_i,$$

where $\mathbb{k}[x_1, \dots, x_n]_i$ is the span of monomials of total degree i .

Example 1.6. Let V be a $\mathbb{k}$ -vector space. The *tensor algebra* on V is

$$T(V) := \bigoplus_{i \geq 0} V^{\otimes i}$$

where $V^{\otimes 0} := \mathbb{k}$. Multiplication is concatenation of tensors:

$$(v_1 \otimes \cdots \otimes v_i)(w_1 \otimes \cdots \otimes w_j) = v_1 \otimes \cdots \otimes v_i \otimes w_1 \otimes \cdots \otimes w_j.$$

Thus $T(V)_i = V^{\otimes i}$. If $V = \mathbb{k}x \oplus \mathbb{k}y$, then $T(V)$ is the free non-commutative polynomial algebra $\mathbb{k}\langle x, y \rangle$. For example, xy and yx are different basis elements of degree 2.

Example 1.7. The exterior algebra on an n -dimensional vector space V is

$$\bigwedge V = \bigoplus_{i=0}^n \bigwedge^i V.$$

It is graded by exterior degree. If $V = \mathbb{k}x \oplus \mathbb{k}y$, then

$$\bigwedge V = \mathbb{k} \oplus \mathbb{k}x \oplus \mathbb{k}y \oplus \mathbb{k}(x \wedge y),$$

with $x^2 = y^2 = 0$ and $xy = -yx$.

Definition 1.8. Let $A = \bigoplus_{i \geq 0} A_i$ and $B = \bigoplus_{i \geq 0} B_i$ be graded $\mathbb{k}$ -algebras. A *homomorphism of graded algebras* $f : A \rightarrow B$ is a $\mathbb{k}$ -algebra homomorphism such that

$$f(A_i) \subset B_i$$

for all $i \geq 0$.

Definition 1.9. A graded algebra $A = \bigoplus_{i \geq 0} A_i$ is called *connected* if $A_0 = \mathbb{k}$. It is called *locally finite* if $\dim_{\mathbb{k}} A_i < \infty$ for all i .

Most examples in the first part of the course will be connected and locally finite. Connectedness means that there is only one copy of the ground field in degree 0. Local finiteness means that each degree can be handled by finite-dimensional linear algebra.

Example 1.10. (1) $\mathbb{k}[x_1, \dots, x_n]$ is connected and locally finite.

(2) $T(V)$ is connected and locally finite if $\dim_{\mathbb{k}} V < \infty$.

(3) $\mathbb{k}[x]$ with $\deg x = 2$ is also a graded algebra, but it is not generated in degree 1.

(4) The Laurent polynomial ring $\mathbb{k}[x, x^{-1}]$ is naturally $\mathbb{Z}$ -graded, but not $\mathbb{Z}_{\geq 0}$ -graded in the above sense.

Graded modules

Definition 1.11. Let $A = \bigoplus_{i \geq 0} A_i$ be a graded algebra. A *graded right A -module* is a right A -module M with a direct sum decomposition

$$M = \bigoplus_{j \in \mathbb{Z}} M_j$$

such that

$$M_j A_i \subset M_{i+j}$$

for all $i \geq 0$ and all $j \in \mathbb{Z}$. An element of M_j is called *homogeneous of degree j* .

We allow negative degrees for modules because shifts are unavoidable. Even if an algebra only has non-negative degrees, its modules may naturally be shifted left or right.

Example 1.12. The algebra A itself is a graded right A -module:

$$A = \bigoplus_{i \geq 0} A_i.$$

More generally, $A^{\oplus r}$ is a graded free module with each basis vector placed in degree 0.

Definition 1.13. Let M be a graded A -module and let $d \in \mathbb{Z}$. The *shift* $M(d)$ is the graded A -module whose underlying A -module is M , but whose grading is

$$M(d)_j = M_{j+d}.$$

This convention means that $A(-d)$ has a generator in degree d . Indeed, $1 \in A_0 = A(-d)_d$.

Example 1.14. Let $A = \mathbb{k}[x]$ with $\deg x = 1$. The graded module $A(-2)$ has basis

$$1, x, x^2, \dots$$

in degrees

$$2, 3, 4, \dots$$

respectively. Thus multiplication by x still raises degree by 1.

Definition 1.15. Let M, N be graded A -modules. A homomorphism $f : M \rightarrow N$ is *homogeneous of degree d* if

$$f(M_j) \subset N_{j+d}$$

for all $j \in \mathbb{Z}$. A *homomorphism of graded modules* means a homogeneous A -module homomorphism of degree 0.

Example 1.16. For $A = \mathbb{k}[x]$, multiplication by x^r gives a homogeneous map

$$A \rightarrow A$$

of degree r . The same multiplication can be regarded as a degree 0 map

$$A(-r) \rightarrow A.$$

This is why shifts appear constantly in graded free resolutions.

Homogeneous ideals and quotients

Definition 1.17. Let $A = \bigoplus_{i \geq 0} A_i$ be a graded algebra. A right ideal $I \subset A$ is **homogeneous** if

$$I = \bigoplus_{i \geq 0} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I .

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \geq 0} A_i / (I \cap A_i)$$

is again a graded algebra.

Example 1.18. In $\mathbb{k}[x, y]$ with the usual grading, the ideal

$$I = (x^2, xy, y^3)$$

is homogeneous. The quotient $A = \mathbb{k}[x, y]/I$ is graded. Its degree 0 part is $\mathbb{k}$, its degree 1 part has basis x, y , and its degree 2 part has basis y^2 . All degrees ≥ 3 vanish.

Example 1.19. The ideal $(x - 1) \subset \mathbb{k}[x]$ is not homogeneous for the usual grading. Indeed $x - 1$ is in the ideal, but its homogeneous components x and -1 are not in the ideal. The quotient $\mathbb{k}[x]/(x - 1) \cong \mathbb{k}$ should not be regarded as a graded quotient of $\mathbb{k}[x]$ by this ideal.

Definition 1.20. Let $A = \bigoplus_{i \geq 0} A_i$ be a connected graded algebra. The ideal

$$A_+ := \bigoplus_{i > 0} A_i$$

is called the **augmentation ideal**. The quotient

$$\mathbb{k}_A := A/A_+$$

is called the **trivial module**.

The trivial module is the simplest graded module over A . Every positive degree element acts by zero. Koszulness will later be a statement about the shape of the minimal graded free resolution of this module.

Hilbert series

Definition 1.21. Let $M = \bigoplus_{j \in \mathbb{Z}} M_j$ be a graded vector space with $\dim_{\mathbb{k}} M_j < \infty$ for all j and $M_j = 0$ for all $j \ll 0$. The *Hilbert series* of M is the formal Laurent series

$$h_M(t) := \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}} M_j) t^j.$$

For a locally finite graded algebra A , we write $h_A(t)$ for its Hilbert series as a graded vector space.

The Hilbert series records the size of each degree. It forgets multiplication, but it is still strong enough to give numerical tests for Koszulness later.

Lemma 1.22. Let M, N be locally finite graded vector spaces.

- (1) If $M \cong N$ as graded vector spaces, then $h_M(t) = h_N(t)$.
- (2) If $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is an exact sequence of graded vector spaces with degree 0 maps, then

$$h_M(t) = h_L(t) + h_N(t).$$

- (3) For any shift $d \in \mathbb{Z}$,

$$h_{M(d)}(t) = t^{-d} h_M(t).$$

Proof All three statements follow by taking the degree j part. For example, $M(d)_j = M_{j+d}$, so

$$h_{M(d)}(t) = \sum_j \dim_{\mathbb{k}}(M_{j+d}) t^j = t^{-d} \sum_i \dim_{\mathbb{k}}(M_i) t^i.$$

□

Example 1.23. For $A = \mathbb{k}[x]$ with $\deg x = 1$, we have $\dim_{\mathbb{k}} A_i = 1$ for all $i \geq 0$. Hence

$$h_A(t) = 1 + t + t^2 + \cdots = \frac{1}{1-t}.$$

For $\mathbb{k}[x_1, \dots, x_n]$ with all variables in degree 1,

$$h_{\mathbb{k}[x_1, \dots, x_n]}(t) = \frac{1}{(1-t)^n}.$$

Indeed, this follows by multiplying the n geometric series, one for each variable.

Example 1.24. Let V be n -dimensional. Then $\dim_{\mathbb{k}} T(V)_i = n^i$, so

$$h_{T(V)}(t) = 1 + nt + n^2 t^2 + \cdots = \frac{1}{1-nt}.$$

On the other hand,

$$h_{\wedge V}(t) = \sum_{i=0}^n \binom{n}{i} t^i = (1+t)^n.$$

Example 1.25. Let $A = \mathbb{k}[x, y]/(x^2, xy, y^3)$. From the basis calculation above,

$$h_A(t) = 1 + 2t + t^2.$$

This series already tells us that A is 4-dimensional as a vector space.

Lemma 1.26. *If V, W are locally finite graded vector spaces, and each graded piece of $V \otimes_{\mathbb{k}} W$ is finite-dimensional, then*

$$h_{V \otimes W}(t) = h_V(t)h_W(t),$$

where $(V \otimes W)_n = \bigoplus_{i+j=n} V_i \otimes W_j$.

Proof The degree n part has dimension

$$\dim_{\mathbb{k}}(V \otimes W)_n = \sum_{i+j=n} \dim_{\mathbb{k}} V_i \dim_{\mathbb{k}} W_j.$$

This is exactly the coefficient of t^n in the product $h_V(t)h_W(t)$. □

Graded duals

Definition 1.27. *Let $V = \bigoplus_{i \in \mathbb{Z}} V_i$ be a locally finite graded vector space. Its **graded dual** is*

$$V^{\vee} := \bigoplus_{i \in \mathbb{Z}} \operatorname{Hom}_{\mathbb{k}}(V_i, \mathbb{k}).$$

We regard $\operatorname{Hom}_{\mathbb{k}}(V_i, \mathbb{k})$ as sitting in degree $-i$.

The ordinary dual $\operatorname{Hom}_{\mathbb{k}}(V, \mathbb{k})$ is a direct product of the duals of the graded pieces. The graded dual is the direct sum. For finite-dimensional graded spaces there is no difference as vector spaces, but for infinite-dimensional examples such as $\mathbb{k}[x]$ the distinction matters.

Example 1.28. *Let $A = \mathbb{k}[x]$ with $\deg x = 1$. The graded dual $A^{\vee}$ has one-dimensional piece in each degree $0, -1, -2, \dots$ and no other degrees. The functional dual to x^i has degree $-i$.*

Left, right, and two-sided duals

The previous definition is enough for connected algebras, where $A_0 = \mathbb{k}$. However, many natural Koszul algebras have degree zero part a semisimple algebra S rather than just $\mathbb{k}$. For example, the path algebra of a quiver has

$$S = \bigoplus_{i \in Q_0} \mathbb{k}e_i$$

in degree 0. In that setting one must keep track of left and right actions.

Definition 1.29. *Let S be a finite-dimensional semisimple $\mathbb{k}$ -algebra.*

(1) *If ${}_S V$ is a finite-dimensional left S -module, define its **right S -dual***

$$V^* := \operatorname{Hom}_S(V, S).$$

It is a right S -module by

$$(fs)(v) := f(v)s.$$

(2) *If W_S is a finite-dimensional right S -module, define its **left S -dual***

$${}^*W := \operatorname{Hom}_{S^{\text{op}}}(W, S).$$

It is a left S -module by

$$(sg)(w) := sg(w).$$

Thus duality reverses sides. This is already familiar over a field: if M is a right A -module, then $\text{Hom}_{\mathbb{k}}(M, \mathbb{k})$ is naturally a left A -module by

$$(af)(m) := f(ma).$$

The same formula explains the general case.

Lemma 1.30. *Let S be finite-dimensional semisimple. For finite-dimensional modules, the evaluation maps*

$$V \rightarrow {}^*(V^*), \quad W \rightarrow ({}^*W)^*$$

are isomorphisms.

Proof Since S is semisimple, every finite-dimensional S -module is a direct summand of a finite direct sum of copies of S . The statement is immediate for S itself, and then follows for finite direct sums and direct summands. $\square$

Definition 1.31. An *S -bimodule* is a vector space V which is both a left and right S -module, with

$$(sv)t = s(vt)$$

for all $s, t \in S$ and $v \in V$. Equivalently, it is a left module over the *enveloping algebra*

$$S^{\text{en}} := S \otimes_{\mathbb{k}} S^{\text{op}}.$$

The bimodule viewpoint is the correct two-sided language. If V is an S -bimodule, then both V^* and *V are again S -bimodules, but the missing action must be defined carefully:

$$(sf)(v) := f(vs) \quad \text{for } f \in V^*,$$

and

$$(gt)(v) := g(tv) \quad \text{for } g \in {}^*V.$$

Together with the actions in the previous definition, these formulas make V^* and *V into bimodules.

Example 1.32. Take $S = \mathbb{k}^r$ with primitive idempotents $e_1, \dots, e_r$. An S -bimodule decomposes as

$$V = \bigoplus_{i,j} e_i V e_j.$$

The summand $e_i V e_j$ should be thought of as arrows whose left end is i and right end is j . Duality reverses the two sides:

$$e_j V^* e_i \cong \text{Hom}_{\mathbb{k}}(e_i V e_j, \mathbb{k}).$$

This is the linear algebra reason that dual quivers reverse arrows.

Example 1.33. If $S = \mathbb{k}^r$ with primitive idempotents $e_1, \dots, e_r$, then an S -module is the same as a direct sum of vector spaces

$$V = V e_1 \oplus \dots \oplus V e_r.$$

Duality over S dualizes each component separately. This is the linear algebra shadow of what happens for quivers with several vertices.

Tensor products also reverse order under duality. For finite-dimensional S -bimodules there are natural isomorphisms

$$W^* \otimes_S V^* \cong (V \otimes_S W)^*, \quad {}^*W \otimes_S {}^*V \cong {}^*(V \otimes_S W).$$

The reversal is not cosmetic: a functional on $V \otimes_S W$ first sees an element of V , then an element of W , while the dual tensor product records the corresponding operations in the opposite order.

Example 1.34. When $S = \mathbb{k}$, this reduces to the usual isomorphism

$$W^* \otimes_{\mathbb{k}} V^* \cong (V \otimes_{\mathbb{k}} W)^*$$

for finite-dimensional vector spaces. The order reversal is usually hidden because vector spaces have no left/right distinction.

Definition 1.35. Let V be a finite-dimensional S -bimodule and let $U \subset V$ be a sub-bimodule. Define the *right orthogonal complement*

$$U^\perp := \{f \in V^* \mid f(U) = 0\},$$

and the *left orthogonal complement*

$${}^\perp U := \{g \in {}^*V \mid g(U) = 0\}.$$

Both are S -sub-bimodules.

This is the notation needed for quadratic duality. If

$$A = T_S(V)/(R), \quad R \subset V \otimes_S V,$$

then the *left quadratic dual* in the sense of BGS is

$$A^! := T_S(V^*)/(R^\perp),$$

where $R^\perp \subset V^* \otimes_S V^* \cong (V \otimes_S V)^*$. There is also a *right quadratic dual*

$${}^!A := T_S({}^*V)/({}^\perp R).$$

When $S = \mathbb{k}$, the two constructions identify canonically with the usual quadratic dual. For general semisimple S , distinguishing them prevents left modules, right modules, and bimodules from being silently interchanged.

2 Lecture 2: Tensor and quadratic algebras

Generated algebras

Let $A = \bigoplus_{i \geq 0} A_i$ be a connected graded algebra. The multiplication maps give linear maps

$$A_1^{\otimes i} \rightarrow A_i, \quad a_1 \otimes \cdots \otimes a_i \mapsto a_1 \cdots a_i.$$

If these maps are surjective for all $i \geq 0$, then every homogeneous element of positive degree can be written as a linear combination of products of degree 1 elements.

Definition 2.1. A connected graded algebra A is *generated in degree 1* if the smallest graded subalgebra of A containing $A_0 = \mathbb{k}$ and A_1 is all of A . Equivalently, the maps

$$A_1^{\otimes i} \rightarrow A_i$$

are surjective for all $i \geq 0$.

Example 2.2. (1) $\mathbb{k}[x_1, \dots, x_n]$ with $\deg x_i = 1$ is generated in degree 1.

(2) $T(V)$ is generated in degree 1, since every tensor is a product of elements of $V = T(V)_1$.

(3) $\bigwedge V$ is generated in degree 1.

(4) $\mathbb{k}[x]$ with $\deg x = 2$ is not generated in degree 1, since $A_1 = 0$ but $A_2 = \mathbb{k}x \neq 0$.

The main construction today explains why tensor algebras are the natural starting point. Every algebra generated in degree 1 is a quotient of a tensor algebra.

Tensor algebras and the universal property

Definition 2.3. Let V be a $\mathbb{k}$ -vector space. The *tensor algebra* of V is

$$T(V) := \bigoplus_{i \geq 0} V^{\otimes i},$$

with $V^{\otimes 0} = \mathbb{k}$ and multiplication given by concatenation.

Thus $T(V)$ consists of finite linear combinations of words in elements of V . It is the algebra with no relations among the generators except those forced by linearity and associativity.

Example 2.4. If $V = \mathbb{k}x$, then

$$T(V) \cong \mathbb{k}[x].$$

There is only one letter, so non-commutativity is invisible. If $V = \mathbb{k}x \oplus \mathbb{k}y$, then

$$T(V) \cong \mathbb{k}\langle x, y \rangle,$$

with basis all words in x and y :

$$1, \quad x, y, \quad x^2, xy, yx, y^2, \quad x^3, x^2y, xyx, \dots$$

Proposition 2.5 (Universal property of the tensor algebra). Let B be a $\mathbb{k}$ -algebra and let $f : V \rightarrow B$ be a $\mathbb{k}$ -linear map. There exists a unique $\mathbb{k}$ -algebra homomorphism

$$\tilde{f} : T(V) \rightarrow B$$

such that $\tilde{f}(v) = f(v)$ for all $v \in V$.

Proof Define

$$\tilde{f}(v_1 \otimes \cdots \otimes v_i) = f(v_1) \cdots f(v_i)$$

on pure tensors and extend linearly. This gives a homomorphism of algebras because concatenation of tensors is sent to multiplication in B . The formula is forced on every product of elements of V , so the homomorphism is unique. $\square$

Corollary 2.6. *Let A be a connected graded algebra generated in degree 1. Then there is a surjective homomorphism of graded algebras*

$$T(A_1) \rightarrow A.$$

Proof Apply the universal property to the inclusion $A_1 \hookrightarrow A$. Since A is generated in degree 1, the induced map is surjective. $\square$

Definition 2.7. *Let A be a connected graded algebra generated in degree 1. Choose $V = A_1$ and write*

$$A \cong T(V)/I$$

for a homogeneous two-sided ideal $I \subset T(V)$. The ideal I is called the [ideal of relations](#).

The vector space V gives the generators. The ideal I tells us which linear combinations of words are declared to be zero. This is the same principle as writing

$$\mathbb{k}[x, y] \cong \mathbb{k}\langle x, y \rangle / (xy - yx).$$

Relations in degree 2

Since $T(V)$ is graded, every homogeneous relation has a degree. Relations in degree 2 are linear combinations of words of length 2, hence elements of $V \otimes V$.

Definition 2.8. *Let V be a finite-dimensional $\mathbb{k}$ -vector space and let*

$$R \subset V \otimes V$$

be a subspace. The algebra

$$A = T(V)/(R)$$

is called a [quadratic algebra](#). Here (R) denotes the two-sided ideal of $T(V)$ generated by R .

In other words, a quadratic algebra is an algebra generated by degree 1 elements whose defining relations are homogeneous relations of degree 2. The word “quadratic” refers to degree, not to commutativity.

Example 2.9. *Let $V = \mathbb{k}x \oplus \mathbb{k}y$. The commutative polynomial ring is*

$$\mathbb{k}[x, y] \cong T(V)/(x \otimes y - y \otimes x).$$

We usually write this as

$$\mathbb{k}[x, y] \cong \mathbb{k}\langle x, y \rangle / (xy - yx).$$

Thus the polynomial ring is quadratic.

Example 2.10. *Let V have basis $x_1, \dots, x_n$. The polynomial ring is*

$$\mathbb{k}[x_1, \dots, x_n] \cong T(V) / (x_i x_j - x_j x_i \mid 1 \leq i < j \leq n).$$

There are $\binom{n}{2}$ independent quadratic commutativity relations.

Example 2.11. Let V have basis $x_1, \dots, x_n$. The exterior algebra is

$$\bigwedge V \cong T(V) / (x_i^2, x_i x_j + x_j x_i \mid 1 \leq i < j \leq n),$$

provided $\text{char } \mathbb{k} \neq 2$. If $\text{char } \mathbb{k} = 2$, the signs require more care; one should define the exterior algebra by the alternating relation $v \otimes v = 0$ for all $v \in V$.

Example 2.12. Let

$$A = \mathbb{k}\langle x, y \rangle / (x^2, xy).$$

Then A is quadratic. A basis of A is given by words in x, y which do not contain x^2 or xy as consecutive subwords. Thus after an x appears, no letter may follow it. The first few basis elements are

$$1, \quad x, y, \quad yx, y^2, \quad y^2x, y^3, \quad \dots$$

Hence

$$h_A(t) = 1 + 2t + 2t^2 + 2t^3 + \dots = 1 + \frac{2t}{1-t}.$$

Example 2.13. The algebra

$$\mathbb{k}[x]/(x^3)$$

with $\deg x = 1$ is generated in degree 1, but it is not quadratic. Its defining relation has degree 3. This algebra is important in ring theory, but it is not in the basic class governed by quadratic duality.

Example 2.14. The algebra

$$\mathbb{k}[x]/(x^2)$$

with $\deg x = 1$ is quadratic:

$$\mathbb{k}[x]/(x^2) \cong T(\mathbb{k}x)/(x \otimes x).$$

As a graded vector space it has basis $1, x$, so

$$h_A(t) = 1 + t.$$

The degree 2 relations of a quotient

Let $A = T(V)/I$ be a connected graded algebra generated in degree 1. The degree 2 part of $T(V)$ is $V \otimes V$. The degree 2 part of the ideal I is

$$I_2 := I \cap (V \otimes V).$$

The quotient map gives

$$A_2 \cong (V \otimes V)/I_2.$$

Thus I_2 is exactly the space of quadratic relations visible in A .

Definition 2.15. Let $A = T(V)/I$ be generated in degree 1. The **quadratic part** of A is

$$qA := T(V)/(I_2).$$

There is a natural surjective homomorphism

$$qA \twoheadrightarrow A.$$

The algebra qA keeps the same generators and the same degree 2 relations as A , but forgets all relations of degree 3 or higher. Thus A is quadratic precisely when $qA \rightarrow A$ is an isomorphism.

Example 2.16. Let $A = \mathbb{k}[x]/(x^3)$ with $\deg x = 1$. Then $V = \mathbb{k}x$ and $I = (x^3)$. Since $I_2 = 0$, we have

$$qA = T(V) \cong \mathbb{k}[x].$$

The natural map $qA \rightarrow A$ is the quotient map $\mathbb{k}[x] \rightarrow \mathbb{k}[x]/(x^3)$, not an isomorphism.

Example 2.17. Let

$$A = \mathbb{k}\langle x, y \rangle / (xy - yx, x^3).$$

Then A is generated in degree 1 but not quadratic. Its quadratic part is

$$qA = \mathbb{k}\langle x, y \rangle / (xy - yx) \cong \mathbb{k}[x, y].$$

The cubic relation $x^3 = 0$ is invisible to qA .

Why quadratic algebras are the right class

Koszul duality begins with the observation that degree 2 relations have a linear dual. If $A = T(V)/(R)$ with $R \subset V \otimes V$, then the dual vector space V^* gives dual generators, while the orthogonal subspace

$$R^\perp \subset V^* \otimes V^*$$

will give the dual quadratic relations. This construction will be made precise next lecture.

For now, note why degree 2 is special. The dual of $V \otimes V$ is naturally $V^* \otimes V^*$ when V is finite-dimensional. Thus a subspace of quadratic relations has an orthogonal complement which is again a space of quadratic relations. There is no similarly simple operation that turns arbitrary cubic and quartic relations into relations of the same kind without adding more structure.

Example 2.18. For the polynomial ring

$$\mathbb{k}[x, y] \cong \mathbb{k}\langle x, y \rangle / (xy - yx),$$

the space of quadratic relations is one-dimensional, spanned by $xy - yx$. The quadratic dual will have two generators x^*, y^* and relations corresponding to the orthogonal complement of this line in

$$V^* \otimes V^*.$$

This produces the exterior algebra on two generators. The full computation will be done in Lecture 3.

Example 2.19. For the exterior algebra

$$\bigwedge(\mathbb{k}x \oplus \mathbb{k}y) \cong \mathbb{k}\langle x, y \rangle / (x^2, y^2, xy + yx),$$

the quadratic dual will be the polynomial ring in two variables. This is the first sign of the duality between symmetric and exterior algebras.

First checks and non-examples

Proposition 2.20. Let $A = T(V)/(R)$ be quadratic, where V is finite-dimensional and $R \subset V \otimes V$. Then A is connected, locally finite, and generated in degree 1.

Proof The tensor algebra $T(V)$ is connected and generated in degree 1. The ideal (R) is homogeneous because it is generated by homogeneous elements of degree 2. Therefore the quotient inherits a connected grading and is generated by the image of V . Since V is finite-dimensional, each $T(V)_i = V^{\otimes i}$ is finite-dimensional, and hence each quotient A_i is finite-dimensional. $\square$

Remark 2.21. The converse is false. A connected locally finite algebra generated in degree 1 need not be quadratic. The example $\mathbb{K}[x]/(x^3)$ is the smallest warning.

Exercise 2.22. Let $A = \mathbb{K}\langle x, y \rangle / (x^2, y^2, xy)$. Find a basis for A_i for $i = 0, 1, 2, 3$, and compute the Hilbert series.

Exercise 2.23. Let $A = \mathbb{K}[x, y] / (x^2, xy, y^2)$ with the usual grading. Show that A is quadratic and compute $h_A(t)$.

Exercise 2.24. Let V be n -dimensional. Show directly that

$$\dim_{\mathbb{K}} \operatorname{Sym}^2(V) = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{K}} \bigwedge^2 V = \frac{n(n-1)}{2}.$$

Interpret these dimensions as the number of independent quadratic monomials in the commutative and anti-commutative cases.